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Digital competencies, student-centered teaching, and teachers' psychological well-being: the mediating role of coping strategies after COVID-19 pandemic

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Psychological well-being among university faculty is a critical factor for sustaining teaching quality and strengthening institutional capacity in higher education. This study examines how digital competencies and student-centered teaching relate to faculty psychological well-being, incorporating the mediating roles of adaptive and maladaptive coping. The sample comprised 233 university instructors from Chilean institutions. Analyses included refined composite measures supported by confirmatory factor analysis and a multiple mediation model. Digital competencies showed a direct positive association with psychological well-being ($\beta = .18$, $p = .010$). Student-centered teaching did not show a direct association with well-being, but it was indirectly associated with it through adaptive coping [B = 0.04, 95% CI (0.01, 0.08)]. In addition, both adaptive coping and lower maladaptive coping were positively associated with psychological well-being. These findings suggest that digital competencies may operate as a direct teaching-related resource linked to faculty well-being, whereas student-centered teaching may contribute indirectly through adaptive coping. The study highlights the relevance of integrating digital training, pedagogical support, and coping-related resources in efforts to strengthen teacher well-being in increasingly digital and participatory higher education environments.

KEYWORDS
digital competencies, higher education, higher education teachers, maladaptive coping, psychological well-being, student-centered teaching

1 Introduction

In higher education, faculty psychological well-being is not only an individual occupational outcome but also a key condition for educational sustainability. Implementing student-centered teaching and digitally mediated instruction requires sustained instructional design effort, socio-emotional work, and ongoing adaptation. Therefore, understanding how pedagogical resources (student-centered teaching orientation) and teaching-related resources (digital competencies) relate to teachers'

psychological well-being is directly relevant to faculty development, instructional quality, and the sustainability of post-pandemic educational innovation (Campos et al. 2018). The psychological well-being of higher education faculty has emerged as a central concern in contemporary research on teaching quality and institutional sustainability (Muñoz-Campos et al. 2018).

Over the last decade—and with particular intensity since the COVID-19 pandemic—university teachers have had to respond to a growing set of demands: mastering digital technologies, maintaining learning-centered teaching standards, adapting to face-to-face, hybrid, or virtual modalities, and sustaining innovative pedagogical practices in uncertain contexts (Expósito and Marsollier 2020; Garrido 2020; Silva-Domínguez and Luque-Villanueva 2022). In Chile, recent evidence indicates that the health crisis and accelerated digitalization have profoundly impacted faculty well-being, increasing perceived stress, symptoms of burnout, and emotional exhaustion (Ávalos et al. 2021; Lücker et al. 2022; Madigan and Kim 2021; Varela et al. 2023; Yurt 2022). Instead of treating well-being as merely the absence of distress, the current study focuses on psychological well-being as positive psychological functioning (i.e., flourishing or eudaimonic well-being). Examining resources and coping strategies may help explain variation in faculty well-being even in situations where stress and burnout are common, because positive well-being and psychological discomfort are conceptually related but distinct aspects (Keyes 2002; Ryff 1989; Bakker and de Vries, 2021).

In this context, two interrelated phenomena are particularly relevant. First, digital transformation has made digital teaching skills an unavoidable professional requirement, revealing training gaps resulting from limited time, scarce resources, and generational differences in technological proficiency (Cáceres and Gómez 2023; Centeno-Caamal and Acuña-Gamboa 2023; Morales et al. 2025; Rodríguez-Fuentes et al. 2023). Second, the consolidation of student-centered approaches, oriented toward active learning, collaboration, and competency-based assessment, has progressively displaced traditional transmissive models (Ayala 2021; Bara and Xhomara 2020; Rögele et al. 2022), imposing new didactic and socio-emotional demands on teaching. Student-centered active strategies may serve as a "double-edged" pedagogical tool: although they can improve student engagement and the perceived meaning of instruction, they may also increase preparation time, coordination demands, and emotional labor, especially in technology-mediated or post-pandemic teaching contexts, complicating their net association with instructor well-being (McLay et al. 2023; Sonnenberg and Rutledge 2024). In these conditions, coping strategies may be particularly important for transforming student-centered practices into sources of positive well-being.

Research shows that when teachers lack the personal resources required to manage these responsibilities, such demands may become risk factors for well-being (Wang et al. 2022). Accordingly, the association between work demands and mental health outcomes is mediated by coping strategies, defined as cognitive and behavioral efforts to manage stressful demands (Budimir et al. 2021; Freire et al. 2020). The use of adaptive strategies—such as planning, seeking support, or positive reinterpretation—has been associated with higher levels of emotional adjustment and more favorable institutional climates for well-being (Bilbao et al. 2025; Deroncele-Acosta et al. 2025). Conversely, maladaptive strategies—such as avoidance or excessive self-criticism—are related to greater psychological distress (Echeburúa and Amor 2019; Schetsche et al. 2023). Because adaptive and maladaptive coping represent qualitatively distinct regulatory pathways with persistent and opposing relationships to teacher adjustment, this study focuses on these two higher-order coping families. While maladaptive strategies—such as denial, avoidance, self-blame, and substance use— tend to hinder recovery and intensify stress, adaptive strategies—such as planning, seeking assistance, and positive reappraisal—mobilize resources and promote active engagement. Both forms of coping are plausible psychological mechanisms within a resource-based framework linking teaching-related resources (digital competencies and student-centered practices) to positive psychological functioning.

In the Chilean context, these tensions are further intensified by gender inequalities, as female academics report greater difficulties balancing work and personal life, along with a heavier emotional burden (Fardella and Schöngut-Grollmus 2025).

Despite progress in this area, a crucial gap still exists: little research has been done on the relationship between faculty psychological well-being and digital competences and student-centered teaching, or how these interactions can be mediated by adaptive and maladaptive coping methods. From an ecological and resource-based perspective, it is reasonable to expect that technological mastery and the adoption of active pedagogical practices function as protective resources that promote job adjustment and professional satisfaction (Jin and Kim 2017; Laca et al. 2011; Navarro Montaño et al. 2022; Soyer 2019). However, their contribution to well-being may depend on specific psychological mechanisms, particularly in contexts where pedagogical innovation is perceived as an additional burden (Echeburúa and Amor 2019; Freire et al. 2020). Descriptive statistics and distributional properties of the main study variables are presented in Table 1.

TABLE 1 Descriptive statistics and distributional properties of the main study variables.

Variable	M	SD	Min	Max	Skewness	Kurtosis
SCAS	4.502	0.523	1.286	5.000	−1.703	5.804
TDC	3.995	0.792	1.455	5.000	−0.815	0.435
ACo	2.046	0.459	0.125	3.000	−0.731	1.741
MCo	2.070	0.507	0.000	3.000	−1.370	2.910
PWB	6.451	0.579	3.375	7.000	−1.685	4.208

SCAS, student-centered active strategies (refined 7-item version); TDC, teachers' digital competencies (22 items); ACo, adaptive coping (16 items); MCo, reverse-coded maladaptive coping (12 items; higher scores indicate lower use of maladaptive coping strategies); PWB, psychological well-being (8 items).

From a theoretical standpoint, this study draws primarily on resource-based frameworks, including the Job Demands–Resources model and Conservation of Resources theory, to examine the relationship between individual and pedagogical resources and faculty well-being through coping mechanisms. The proposed model focuses on individual-level relationships consistent with the available data, while acknowledging broader ecological implications for teaching practice.

A mediation model grounded in Conservation of Resources theory and the Job Demands–Resources framework was tested. According to this model, teachers' digital competencies (TDC) and student- centered active strategies (SCAS) operate as pedagogical and personal resources associated with higher psychological well-being (PWB), partly through coping mechanisms (Bakker and Demerouti 2017; Demerouti et al. 2001; Hobfoll 1989). Accordingly, the following hypotheses were proposed: (H1) TDC would be positively associated with PWB; (H2) SCAS would be positively associated with PWB; (H3) TDC and SCAS would be positively associated with adaptive coping; (H4) adaptive coping would be positively associated with PWB; and (H5) adaptive coping would partially mediate the associations between TDC, SCAS, and PWB. The moderating role of maladaptive coping in occupational contexts was examined on an exploratory basis due to inconsistent findings in prior research (H6).

The objective of this study is to analyze the direct and indirect effects of digital competencies and student-centered teaching on the psychological well-being of university faculty, considering the mediating role of adaptive and maladaptive coping strategies. The study seeks to provide empirical evidence in an underexplored area and to contribute conceptual insights relevant to the design of teacher development policies and institutional programs aimed at strengthening professional well-being. Within this framework, it is expected that technological mastery and the adoption of active teaching practices will be positively related to well-being and that these relationships will operate, at least in part, through the use of more adaptive coping strategies.

2 Method

2.1 Type of study

A quantitative, non-experimental, cross-sectional study with a correlational-explanatory scope was conducted. This design is relevant for analyzing relationships between variables and estimating mediation models at a single point in time. Although it does not allow causality to be established, it is suitable for identifying potential mechanisms that link teaching resources—digital skills and student-centered teaching—with the psychological well-being of teachers.

2.2 Participants

The sample consisted of 233 higher education instructors from Chilean institutions, selected through non-probabilistic convenience sampling. Participants were recruited through institutional mailing lists and professional academic networks and completed an anonymous online survey after providing informed consent. The sample included 110 women (47.21%) and 123 men (52.79%). Teaching experience was heterogeneous, including both early-career and more experienced instructors. Most participants worked in private institutions (66.95%), while 33.05% were affiliated with public institutions. In terms of disciplinary affiliation, the sample was composed mainly of instructors from Health (42.49%) and Education (29.18%), with the remaining participants distributed across other academic and professional areas.

2.3 Ethical considerations

The study was approved by an institutional ethics committee. All participants signed an informed consent form, ensuring voluntariness, anonymity, and confidentiality in the treatment of data. No information was collected that would allow for the personal identification of teachers.

2.4 Instruments

Five self-report scales were used. For all of them, structural validity was evaluated through Confirmatory Factor Analysis (CFA). The final models met conventional global fit criteria (CFI/TLI $\geq$ .90; RMSEA $\leq$ .08; SRMR $\leq$ .08) (Hu and Bentler, 1999). Internal consistency was estimated using Cronbach's alpha (α), with values ranging from.81 to.96. Unless explicitly stated otherwise, higher scores indicate higher levels of the construct being assessed. For subsequent analyses, scale scores were operationalized as the mean of the retained items for each scale.

(a) Student-centered active strategies (SCAS)

Student-centered active strategies (SCAS) were initially assessed using the 16 items specified for this construct. Responses were rated on a 5-point Likert scale ranging from 1 (strongly disagree) to 5 (strongly agree), with higher scores indicating stronger endorsement of student-centered teaching practices. The initial item pool was designed to capture active, constructivist, and conceptually oriented instructional practices. Illustrative items from the final retained version include: "I strive for students to apply course content to real-life situations," "Students work actively in constructing their own knowledge," and "Learning in this course requires students to interpret information critically." Additional retained items reflected practices such as encouraging students to question their ideas, express their own views, connect concepts, and demonstrate deep understanding through assessment. To examine factorial validity, an independent confirmatory factor analysis (CFA) was conducted, beginning with the full 16-item set. Because the initial one-factor solution showed inadequate fit, a parsimonious refinement process was undertaken through sequential item removal based on low factor loadings and overall model fit, resulting in a final 7-item unidimensional solution. The final model showed adequate fit to the data, $\chi^2(14) = 34.64$, RMSEA = .080, CFI = .958, TLI = .938, and SRMR = .045. For subsequent analyses, SCAS was operationalized as the mean of the retained items (7). Internal consistency was satisfactory (Cronbach's α = .831).

b. Teacher digital competencies (TDC)

Teacher digital competencies were assessed using the DigCompEdu Check-In scale, developed by Cabero-Almenara et al. (2020). This instrument consists of a 22-item Likert-type scale with response options ranging from 1 to 5 and covers key dimensions of digital teaching competence, including the use of digital technologies for institutional communication, participation in online communities of teaching practice, ongoing professional development, selection or creation of digital resources, planning of digital tasks aligned with learning objectives, technology-mediated collaborative learning, digitally supported assessment, and the design of inclusive and accessible activities.

llustrative items include: "Use of digital technologies for institutional communication," "Participation in online communities of teaching practice," and "Critical evaluation of one's own practice using digital technologies".

The initial one-factor CFA indicated moderate fit. To improve model specification, residual covariances were added between items with nearly duplicated content, without requiring item removal. The final one-factor model with 22 items showed good fit: $\chi^2(210) = 494.39$, RMSEA = .08, CFI = .92, TLI = .92, SRMR = .05. Factor loadings were high and homogeneous. Reliability was excellent ($\alpha = .96$). The TDC score was operationalized as the mean of the 22 items.

c. Adaptive coping (ACo)

Adaptive coping (ACo) was assessed using 16 items from the Brief COPE (COPE-28; Carver, 1997), translated into Spanish by Consuelo Morán and Dionisio Manga. Responses were rated on a 4-point scale ranging from 0 (never) to 3 (almost always). The selected items captured adaptive coping tendencies related to active coping, planning, cognitive reframing, acceptance, social support, humor, and religion/spirituality. Illustrative items include: "I try to get someone to help me or advise me about what to do," "I concentrate my efforts on doing something about the situation I'm in," and "I accept the reality of what has happened." Prior to confirmatory factor analysis (CFA), an exploratory factor analysis using principal axis factoring was conducted to examine dimensionality. Results showed a dominant first factor (eigenvalue = 4.10) accounting for 25.59% of the total variance, although the second factor showed an eigenvalue slightly above 1 (1.09), suggesting some residual multidimensionality within the item set. A CFA was then conducted, and the initial one-factor solution showed limited fit. Based on modification indices and loading patterns, residual covariances were specified between highly similar items, and four items were removed. The final 12-item one-factor model showed adequate fit, $\chi^2(55) = 126.25$, RMSEA = .07, CFI = .91, TLI = .91, SRMR = .05, and satisfactory internal consistency (Cronbach's $\alpha = .81$). For subsequent analyses, ACo was operationalized as the mean of the 12 retained items.

d. Maladaptive coping (MCo)

Maladaptive coping (MCo) was assessed using 12 items from the Brief COPE (COPE-28; Carver, 1997), translated into Spanish by Consuelo Morán and Dionisio Manga. Responses were rated on a 4-point scale ranging from 0 (never) to 3 (almost always). The selected items captured maladaptive coping tendencies such as avoidance, denial, self-criticism, emotional venting, and substance use. Illustrative items include: "I use alcohol or other drugs to help me get through it," "I give up trying to deal with it," and "I blame myself for what happened." Prior to confirmatory factor analysis (CFA), an exploratory factor analysis using principal axis factoring was conducted to examine dimensionality. Results indicated a dominant first factor (eigenvalue = 3.92) accounting for 32.68% of the total variance, whereas the second factor showed an eigenvalue below 1 (0.80), supporting the use of a higher-order composite maladaptive coping index. For analytic consistency across study variables, items were reverse-coded so that higher scores reflected lower use of maladaptive coping strategies. A CFA was then conducted with the full 12-item set. Because some items showed closely related semantic content, theoretically justified residual covariances were specified to improve model fit, without removing any items. The final one-factor model showed adequate fit, $\chi^2(55) = 124.66$, RMSEA = .07, CFI = .93, TLI = .93, SRMR = .06, and satisfactory internal consistency (Cronbach's $\alpha = .83$). For subsequent analyses, MCo was operationalized as the mean of the 12 items.

e. Psychological well-being (PWB)

Psychological well-being was measured using 8 Likert-type items (1–7) reflecting facets of teachers' well-being, including purpose in life, supportive relationships, engagement in daily activities, contribution to others' well-being, perceived competence, positive life evaluation, optimism, and perceived respect. Items included statements such as: "I actively contribute to the happiness and well-being of others" and "I am competent and capable in the activities that are important to me." The one-factor CFA indicated good fit: $\chi^2(21) = 52.08$, RMSEA = .08, CFI = .96, TLI = .96, SRMR = .06.

All factor loadings were high and statistically significant, supporting the unidimensionality of the construct. Internal consistency was high ($\alpha = .86$). The PWB index was computed as the mean of the 8 items.

2.5 Procedure

The questionnaires were administered online using an institutional form in Microsoft Forms. Participation was voluntary and without incentives. Before analysis, missing data were examined. We employed the complete-case sample (listwise deletion) with no missing values on the five composite indices for the mediation models, resulting in an analytic sample of $N = 233$.

2.6 Statistical analyses

Confirmatory factor analysis (CFA) was used to evaluate the factorial validity of each study scale. Model fit was assessed using chi-square (χ^2), the root mean square error of approximation (RMSEA), the comparative fit index (CFI), the Tucker–Lewis index (TLI), and the standardized root mean square residual (SRMR). When necessary, theoretically justified refinements were implemented, including sequential item removal in the case of SCAS and residual covariances between semantically overlapping items in the case of TDC, ACo, and MCo.

Prior to CFA, exploratory factor analyses were conducted for the coping scales to examine dimensionality. For ACo, principal axis factoring indicated a dominant first factor (eigenvalue = 4.10) accounting for 25.59% of the total variance, although the second factor showed an eigenvalue slightly above 1 (1.09), suggesting some residual multidimensionality. For MCo, the first factor showed an eigenvalue of 3.92 and accounted for 32.68% of the total variance, whereas the second factor showed an eigenvalue below 1 (0.80), supporting the use of a higher-order composite maladaptive coping index.

To assess the potential influence of common method bias, a full collinearity assessment was conducted following Kock (2015). Variance inflation factors (VIF) were estimated for the composite scores of the constructs included in the model (TDC, SCAS, ACo, MCo, and PWB). Full collinearity VIF values ranged from 1.08 to 1.34, all of them below the conservative threshold of 3.3.

Descriptive statistics, Cronbach's alpha coefficients, and Pearson correlations were computed for all study variables. Direct and indirect associations were then estimated using a regression-based path analysis with observed composite scores, specifying teachers' digital competencies (TDC) and student-centered active strategies (SCAS) as predictors, adaptive coping (ACo) and maladaptive coping (MCo) as mediators, and psychological well-being (PWB) as the outcome. Indirect effects were estimated using nonparametric bootstrapping with 2,000 resamples and were considered statistically significant when the 95% confidence intervals did not include zero.

3 Results

3.1 Descriptive statistics

Descriptive results suggest a generally favorable self-reported profile among participating teachers. Mean scores were located in the upper range for student-centered active strategies and psychological well-being, while digital competencies also showed comparatively elevated levels. This pattern indicates that the sample tended to perceive itself as pedagogically active, digitally competent, and psychologically well-adjusted. In parallel, the coping indicators were consistent with a predominantly adaptive profile, as adaptive coping reached moderate-to-high levels and the reverse-coded maladaptive coping index suggested a relatively low reliance on avoidant or dysfunctional strategies. From a distributional standpoint, the negative skewness observed across most variables—particularly for SCAS and PWB—indicates clustering toward the upper end of the response scales, which is consistent with potential ceiling tendencies and reduced variability in these constructs. This restricted dispersion is analytically relevant because it may attenuate the magnitude of bivariate associations and partially account for modest effect sizes in the regression-based mediation model.

3.2 Correlations among the constructs

The bivariate correlations were broadly consistent with the proposed model. Teachers' digital competencies were positively associated with student-centered active strategies ($r = .41$), adaptive coping ($r = .22$), and psychological well-being ($r = .21$). Student-centered active strategies were also positively associated with adaptive coping ($r = .26$), whereas their association with psychological well-being was small ($r = .11$). In turn, adaptive coping showed a moderate negative correlation with maladaptive coping ($r = -.42$). Taken together, these findings suggest that pedagogical and technological resources were more strongly related to adaptive coping and well-being than to lower use of maladaptive coping strategies (see Table 2).

3.3 Regression-based mediation model

The mediation model enabled an integrated analysis of the direct and indirect associations between teachers' digital competencies, student-centered active strategies, and psychological well-being, considering the mediating roles of adaptive and maladaptive coping. The model was estimated using observed composite scores derived from the refined measurement solutions described above. Accordingly, CFA was used to validate the measurement of each scale, whereas the direct and indirect effects reported below were estimated through regression-based path analysis. The results of the multiple mediation model, including direct and indirect effects, are presented in Table 3.

3.3.1 Predictors of coping

The structural model first examined the extent to which teachers' digital competencies and student-centered active strategies were associated with coping styles. In the equation predicting adaptive coping, both predictors jointly explained

TABLE 2 Pearson correlation matrix between constructs.

Variable	TDC	SCAS	ACo	MCo	PWB
TDC	1.00	.41**	.22**	−.08	.21**
SCAS		1.00	.26**	−.09	.11
ACo			1.00	−.42**	.12
MCo				1.00	.11
PWB					1.00

TDC, teachers' digital competencies; SCAS, student-centered active strategies; ACo, adaptive coping; MCo, maladaptive coping; PWB, psychological well-being.
**$p < .01$.

TABLE 3 Results of the structural mediation model (path analysis with observed variables).

Dependent variable	Predictor	B	β	p	R^2
ACo	TDC	0.08	.13	.055	
	SCAS	0.18	.21	.003	.08
MCo	TDC	−0.03	−.05	.504	
	SCAS	−0.07	−.07	.347	.01
PWB	TDC	0.13	.18	.010	
	SCAS	0.01	.01	.920	
	ACo	0.20	.16	.034	
	MCo	0.22	.19	.007	.08

approximately 8% of the variance ($R^2 = .08$). Student-centered active strategies showed a positive and statistically significant association with adaptive coping (SCAS → ACo: B = 0.18, $\beta = .21$, $p = .003$), whereas the association between teachers' digital competencies and adaptive coping was positive but did not reach conventional levels of statistical significance (TDC → ACo: B = 0.08, $\beta = .13$, $p = .055$). These findings indicate that teachers reporting stronger endorsement of student-centered active strategies also tended to report greater use of adaptive coping strategies, while digital competencies showed a weaker and only marginal association with this coping dimension.

In contrast, when maladaptive coping was used as the dependent variable, the model explained little variance ($R^2 = .01$), and neither predictor showed a statistically significant association (TDC → MCo: B = −0.03, $\beta = -.05$, $p = .504$; SCAS → MCo: B = −0.07, $\beta = -.07$, $p = .347$). Thus, within the present model, neither digital competencies nor student-centered teaching practices were robustly associated with lower use of maladaptive coping strategies.

3.3.2 Predictors of psychological well-being

The regression model predicting psychological well-being from teachers' digital competencies, student-centered active strategies, adaptive coping, and maladaptive coping explained approximately 8% of the variance ($R^2 = .08$). Although statistically significant, this proportion of explained variance is modest, suggesting that additional individual, organizational, and contextual factors not included in the present model likely contribute to differences in teachers' psychological well-being.

Teachers' digital competencies showed a positive and statistically significant direct association with psychological well-being (TDC → PWB: B = 0.13, $\beta = .18$, $p = .010$). In contrast, student-centered active strategies did not show a meaningful direct association with well-being (SCAS → PWB: B = 0.01, $\beta = .01$, $p = .920$). Both adaptive coping (ACo → PWB: B = 0.20, $\beta = .16$, $p = .034$) and lower maladaptive coping (MCo → PWB: B = 0.22, $\beta = .19$, $p = .007$) were positively associated with psychological well-being. Taken together, these results indicate that teachers with stronger digital competencies and more adaptive coping profiles tended to report higher levels of well-being, whereas the contribution of student-centered teaching operated primarily through indirect rather than direct pathways.

3.3.3 Indirect effects

Indirect effects were estimated using bootstrapping with 2,000 resamples. For teachers' digital competencies, the indirect effect through adaptive coping was not statistically significant [via ACo: B = 0.02, 95% CI (−0.00, 0.04)], as the confidence interval included zero. Likewise, the indirect effect through maladaptive coping was not significant [via MCo: B = −0.01, 95% CI (−0.03, 0.01)]. Given that the direct association between digital competencies and psychological well-being remained positive and statistically significant, these findings suggest that this relationship was primarily direct rather than mediated by coping mechanisms.

By contrast, student-centered active strategies showed a positive and statistically significant indirect association with psychological well-being through adaptive coping [via ACo: B = 0.04, 95% CI (0.01, 0.08)]. This pattern indicates that greater endorsement of student-centered teaching was associated with increased reliance on adaptive coping, which in turn was linked to higher psychological well-being. The indirect effect through maladaptive coping was not significant [via MCo: B = −0.01, 95% CI (−0.04, 0.02)], and the direct association between SCAS and PWB remained non-significant. Overall, the structural model suggests that digital competencies functioned as a direct resource associated with well-being, whereas student-centered active strategies contributed indirectly through adaptive coping.

4 Discussion

The present study examined how digital competencies and student-centered teaching were associated with university teachers' psychological well-being, considering the roles of adaptive and maladaptive coping strategies. Taken together, the findings partially support the proposed theoretical model and align with recent research on teacher well-being in contexts marked by accelerated digitalization and post-pandemic pedagogical reorganization. More specifically, digital competencies showed a direct positive association with psychological well-being, whereas student-centered teaching did not show a direct association with well-being but was indirectly linked to it through adaptive coping. In turn, both adaptive coping and lower maladaptive coping were positively associated with well-being, although maladaptive coping did not operate as a significant mediator. Although the model was informed by JD-R theory, job demands were not directly measured in the present study. Accordingly, our application of JD-R is partial and focused on individual and pedagogical resources; future research should incorporate direct indicators of demands (e.g., workload, role conflict, administrative pressure) and organizational resources within multilevel JD-R designs.

4.1 Relationship between digital competencies and psychological well-being

Digital competencies showed a positive and significant direct association with psychological well-being, even when controlling for teaching practices and coping strategies. This finding converges with prior evidence indicating that technological proficiency reduces uncertainty, operational overload, and stress associated with hybrid or ICT-mediated teaching (König et al. 2020). In the Latin American context, recent studies have shown that digital competence is associated with both ICT-based educational innovation and lower levels of burnout and work-related stress (Deroncele-Acosta 2024, 2025).

From the perspective of Conservation of Resources theory (Hobfoll 1989), digital competencies can be understood as a structural resource that may strengthen teachers' sense of self-efficacy, promote work efficiency, and reduce strain associated with continuous adaptation. In the present study, their association with well-being did not require significant indirect pathways through coping, which is consistent with the absence of a significant mediated effect. Accordingly, institutional investment in digital training may contribute not only to

strengthening pedagogical practices but also to supporting teacher well-being, as suggested by recent European and Latin American research (Marpa 2021).

4.2 Student-centered teaching: lack of direct effect and presence of indirect effects

Unlike digital competencies, student-centered teaching did not show a direct association with psychological well-being. This pattern is consistent with research indicating that implementing active methodologies—particularly in technology-mediated environments—may increase technical workload, pedagogical uncertainty, and the emotional complexity of the teaching role (McLay et al. 2023; Sonnenberg et al. 2024). In the present sample, the high mean levels observed for SCAS also suggest restricted variance, which may have reduced the likelihood of detecting stronger direct associations with well-being. This interpretation is further supported by the negative skewness observed for the construct, which is compatible with potential ceiling effects in self-reported student-centered practices.

At the same time, the structural model identified a significant indirect association between student-centered teaching and psychological well-being through adaptive coping. This finding suggests that a stronger orientation toward active and participatory teaching may be linked to greater use of coping strategies such as planning, positive reappraisal, and help-seeking, which in turn are associated with better well-being. In this sense, student-centered teaching may not operate as a direct source of well-being, but rather as a pedagogical context that is more likely to be beneficial when accompanied by adaptive self-regulation resources. This interpretation is consistent with literature showing that flexible and strategic coping with classroom demands is associated with higher levels of teacher well-being, particularly in contexts of pedagogical change (Nwoko et al. 2023; Stockinger et al. 2025).

4.3 Adaptive and maladaptive coping as predictors of well-being

Consistent with the Job Demands–Resources (JD-R) model, coping may function as a personal resource that shapes how teaching-related demands and resources are linked to well-being (Bakker and Demerouti 2017). In the present study, both adaptive coping and lower maladaptive coping were positively associated with psychological well-being in the structural model, aligning with international evidence linking effective emotional regulation with better mental health among teachers (Dreer 2022; García-Carmona et al. 2022; Shvay et al. 2025; Sokal et al. 2020; Vargas Rubilar and Oros 2021). However, this JD-R–based interpretation should be understood as a partial evaluation focused on individual and pedagogical resources and coping, as the current study did not directly assess organizational resources (e.g., administrative support) or job demands (e.g., workload, role conflict). To comprehensively examine the JD-R process in higher education settings, future research should incorporate indicators of job demands and multilevel organizational resources (Bakker and Demerouti 2017; Demerouti et al. 2001).

Maladaptive coping did not mediate the relationships between teaching-related variables and psychological well-being. This finding suggests that maladaptive strategies—such as avoidance, self-criticism, or denial—are more strongly influenced by personal and contextual factors (e.g., job stability, institutional climate, or intolerance of uncertainty), as reported in recent research on teacher well-being (Kim and Asbury 2020; Shvay et al. 2025; Vargas Rubilar and Oros 2021). This pattern implies that reducing maladaptive coping requires broader interventions that extend beyond pedagogical or technological training alone.

4.4 Synthesis and contributions of the study

The findings support two complementary pathways linking teaching-related resources to university teachers' psychological well-being. The first is a direct pathway, in which digital competencies emerge as a personal and professional resource positively associated with well-being independently of coping. The second is an indirect pathway, in which student-centered teaching is associated with well-being through stronger adaptive coping. Together, these pathways provide a more integrated understanding of teacher well-being by linking pedagogical practices, digital preparedness, and coping processes within a single explanatory framework.

Beyond confirming specific hypotheses, the study contributes to current knowledge by showing that well-being is not solely associated with technological skills or pedagogical orientation in isolation, but also with the ways teachers regulate the demands associated with pedagogical change. In methodological terms, the study also contributes by refining the measurement of student-centered active strategies through CFA and by examining the dimensionality of both coping constructs through the combined use of EFA and CFA. Whereas maladaptive coping showed clearer support for a higher-order composite index, adaptive coping retained some residual multidimensionality, consistent with the conceptual breadth of the construct.

4.5 Implications for training and educational management

The results offer relevant implications for higher education institutions. First, they suggest that investment in digital training may be associated not only with stronger teaching performance but also with better psychological well-being among instructors, which is particularly relevant in contexts of continuous curricular innovation. Second, they indicate that student-centered teaching initiatives may be more beneficial when accompanied by institutional support for adaptive coping, since it is this pathway—and not the pedagogical orientation alone—that was linked to well-being in the present model. In practical terms, institutions could incorporate coping-related competencies into faculty development, establish peer-based support structures such as mentoring and communities of practice, and provide protected time, instructional design

support, and accessible technical assistance to reduce implementation overload.

Second, the findings point to the need to accompany student-centered teaching initiatives with programs explicitly aimed at strengthening adaptive coping, since it is this component—and not the pedagogical practice by itself—that was associated with well-being in the present model. Finally, the findings underscore the importance of designing institutional support policies that address factors influencing maladaptive coping, such as working conditions, academic workloads, and organizational environment.

Although the proportion of variance explained in psychological well-being was modest, this result should be interpreted within the broader conceptual understanding of well-being as a complex and multifactorial construct. Teachers' psychological well-being is shaped by a wide constellation of individual, interpersonal, organizational, and contextual factors —including workload, institutional climate, leadership, job security, and role demands—that were not directly modeled in the present study. Accordingly, the current findings should be understood as identifying specific pedagogical and coping-related pathways within a broader explanatory landscape rather than providing a comprehensive account of well-being determinants.

4.6 Limitations and future lines of research

This study has several limitations that should be considered. First, given the cross-sectional design, all findings should be interpreted as associations rather than evidence of causal effects. Although mediation analyses were conducted, cross-sectional mediation models represent statistical decompositions of concurrent relationships rather than demonstrations of causal processes or temporal directionality, as mediation is inherently temporal. Accordingly, the indirect pathways identified in this study should be regarded as exploratory and hypothesis-generating. Prior methodological research indicates that cross-sectional mediation estimates may be biased relative to longitudinal mediation effects (Cole and Maxwell 2003; Maxwell and Cole 2007), underscoring the need for longitudinal or cross-lagged designs to establish temporal ordering and causal inference.

Second, the use of non-probabilistic convenience sampling limits external validity and generalizability beyond the participating Chilean institutions. Additionally, all constructs were measured via self-report instruments administered within the same survey context, which may introduce shared method variance and social desirability bias. To evaluate potential common method bias, we applied a full collinearity VIF assessment (Kock, 2015); full collinearity VIF values ranged from 1.08 to 1.34, all of them below the conservative threshold of 3.3, suggesting that common method bias is unlikely to have materially inflated the observed relationships. Nevertheless, future research should incorporate multi-source indicators (e.g., peer or student ratings, behavioral measures, or administrative data) and apply additional procedural and statistical remedies to further mitigate method-related bias (Podsakoff et al. 2003).

Third, several variables exhibited high mean levels—particularly psychological well-being and student- centered teaching—indicating potential ceiling effects and restricted variance. This pattern may attenuate correlations and partially account for modest effect sizes observed in the structural model. Future studies would benefit from more heterogeneous samples and measurement strategies capable of capturing greater variability in key constructs. Also, an additional measurement-related consideration concerns adaptive coping, as the preliminary EFA supported a dominant first factor, although some residual multidimensionality was also observed, in line with the conceptual breadth of the construct. The refined CFA solution nonetheless showed adequate fit, and future studies could explore this measure using multidimensional or higher-order modeling approaches.

Despite these limitations, the findings contribute to the literature by integrating digital competencies, pedagogical practices, and coping processes within a theoretically grounded explanatory framework, offering a useful basis for informing institutional policies and academic development initiatives in higher education.

5 Conclusion

This study provides evidence that digital competencies and student-centered teaching are associated with university teachers' psychological well-being through different pathways. Digital competencies showed a direct positive association with well-being, whereas student-centered teaching did not show a direct association with psychological well-being but was indirectly linked to it through adaptive coping. In addition, both adaptive coping and lower maladaptive coping were positively associated with well-being. Although the explained variance was modest, these findings support the relevance of pedagogical and coping-related resources for understanding teacher well-being in higher education.

The findings reinforce the importance of addressing teacher well-being as a multidimensional issue situated within ongoing processes of pedagogical and technological change. In contexts marked by digital transformation, active teaching demands, and increasing institutional pressure to innovate, digital preparedness, student-centered practices, and coping resources appear to represent interrelated dimensions of teachers' professional experience. From this perspective, institutional efforts to strengthen faculty well-being may benefit from combining digital training, pedagogical support, and resources that foster adaptive coping in response to the demands of contemporary teaching.

At the same time, the results should be interpreted within a broader multicausal framework. Teachers' psychological well-being is likely associated with additional organizational and contextual conditions not included in the present model, such as workload, leadership, administrative support, and employment conditions. Future research should incorporate these predictors, together with longitudinal and multilevel designs, in order to provide a more comprehensive understanding of teacher well-being in contemporary higher education.

Data availability statement

The raw data supporting the conclusions of this article will be made available by the authors, without undue reservation.

Ethics statement

Ethical approval was not required for the studies involving humans because Ethical approval was not required for this study in accordance with national legislation and institutional requirements. The study involved adult participants only, data were collected through anonymous self-report questionnaires, and no sensitive personal information was obtained. All procedures were conducted in accordance with the ethical principles of the Declaration of Helsinki. Participation was voluntary, and informed consent was obtained from all participants prior to data collection. The studies were conducted in accordance with the local legislation and institutional requirements. The participants provided their written informed consent to participate in this study.

Author contributions

CC-C: Writing – original draft, Writing – review & editing. DR: Conceptualization, Funding acquisition, Investigation, Validation, Writing – review & editing. AP: Data curation, Investigation, Writing – review & editing. SF-U: Conceptualization, Formal analysis, Investigation, Validation, Writing – review & editing. JE-Á: Conceptualization, Investigation, Writing – review & editing. WF: Data curation, Investigation, Writing – review & editing.

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Conflict of interest

The author(s) declared that this work was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

Generative AI statement

The author(s) declared that generative AI was used in the creation of this manuscript. The authors verify and take full responsibility for the use of generative artificial intelligence in the preparation of this manuscript. Generative AI tools were used exclusively for language editing, stylistic refinement, and minor corrections related to grammar, clarity, and consistency. No generative AI was used to create original scientific content, analyze data, interpret results, or generate conclusions. All substantive intellectual contributions, data analyses, and interpretations are the sole responsibility of the authors.

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References

Ayala, R. (2021). Un zoom a la educación virtual: biopolítica y aprendizaje centrado en el estudiante. *Educ. Méd.* 22 (3), doi: 10.1016/j.edumed.2021.01.004

Ávalos, B., Cavada, P., Pardo, M., and Sotomayor, C. (2021). La profesión docente en Chile: cambios, tensiones y desafíos. *Estudios Pedagógicos*. 47 (1), 1–18.

Bakker, A. B., and de Vries, J. D. (2021). Job demands–resources theory and self-regulation: new explanations and remedies for job burnout. *Anxiety. Stress. Coping.* 34 (1), 1–21. doi: 10.1080/10615806.2020.1797695

Bakker, A. B., and Demerouti, E. (2017). Job demands–resources theory: taking stock and looking forward. *J. Occup. Health. Psychol.* 22 (3), 273–285. doi: 10.1037/ocp0000056

Bara, G., and Xhomara, N. (2020). The effect of student-centered teaching and problem- based learning on academic achievement in science. *J. Turk. Sci. Educ.* 17 (2), 180–198. doi: 10.36681/tused.2020.20

Bilbao, M., López, V., Torres-Vallejos, J., Ascorra, P., and Páez, D. (2025). Adaptación de la escala de bienestar social para profesorado y personal escolar. *Rev. Psicol. (PUCP)* 43 (1), 29–52. doi: 10.18800/psico.202501.002

Budimir, S., Probst, T., and Pieh, C. (2021). Coping strategies and mental health during COVID-19 lockdown. *J. Mental Health* 30 (2), doi: 10.1080/09638237.2021.1875412

Cabero-Almenara, J., Romero-Tena, R., and Palacios-Rodríguez, A. (2020). Evaluation of teacher digital competence frameworks through expert judgement: the use of the DigCompEdu Check-In questionnaire. *Sustainability*. 12 (22), 9156.

Campos, M., Elsa, M., González, A. F., and Jacott, L. (2018). Teachers' subjective well-being and satisfaction with life. *REICE* 16 (1), doi: 10.15366/reice2018.16.1.007

Cáceres, D., and Gómez, I. (2023). La transformación digital en latinoamérica, una necesidad y oportunidad para mejorar la educación superior. *Investig. Pensamiento Crít.* 11 (3), doi: 10.37387/ipc.v11i3.362

Carver, C. S. (1997). You want to measure coping but your protocol's too long: Consider the Brief COPE. *Int. J. Behav. Med.* 4 (1), 92–100. doi: 10.1207/s15327558ijbm0401_6

Centeno-Caamal, R., and Acuña-Gamboa, L. (2023). Competencias digitales docentes y formación continua: una propuesta desde el paradigma cualitativo. *RELATEC* 22 (2), 119–134. doi: 10.17398/1695-288X.22.2.119

Cole, D. A., and Maxwell, S. E. (2003). Testing mediational models with longitudinal data: questions and tips in the use of structural equation modeling. *J. Abnorm. Psychol.* 112 (4), 558–577. doi: 10.1037/0021-843X.112.4.558

Demerouti, E., Nachreiner, F., Bakker, A. B., and Schaufeli, W. B. (2001). The job demands–resources model of burnout. *J. Appl. Psychol.* 86, 3. doi: 10.1037/0021-9010.86.3.499

Deroncele-Acosta, A., Rojas-Vistorte, A. O., Sartor-Harada, A., Ulloa-Guerra, O., López-Mustelier, R., and Cruzata-Martínez, A.. (2024). Positive mental health of Latin American university professors: a scientific framework for intervention and improvement. *Heliyon*. 10 (2).

Deroncele-Acosta, A., Norabuena-Figueroa, R. P., Mustelier, R. L., Cao, E. R., and Sotomayor-Solís, P. (2025). Escala de bienestar del empleado docente en América Latina (EWBTS-LATAM). *SAGE Open*. 15 (3). doi: 10.1177/21582440251379089

Dreer, B. (2022). Teacher well-being: Investigating the contributions of school climate and job crafting. *Cogent Educ.* 9 (1), 2044583. doi: 10.1080/2331186X.2022.2044583

Echeburúa, E., and Amor, P. J. (2019). Memoria traumática: estrategias de afrontamiento adaptativas e inadaptativas. *Ter. Psicol.* 37 (1), doi: 10.4067/S0718-48082019000100071

Expósito, C. D., and Marsollier, R. G. (2020). Virtualidad y educación en tiempos de COVID-19: un estudio empírico en Argentina. *Educ. Hum.* 22 (39), doi: 10.17081/eduhum.22.39.4214

Fardella Cisternas, C., and Schöngut-Grollmus, N. (2025). Cuerpos en rendimiento: tecnologías, autoseguimiento y conciliación trabajo-familia en mujeres académicas de alto desempeño. *Psicoperspectivas* 24 (2), doi: 10.5027/psicoperspectivas-Vol24-Issue2-fulltext-3450

Freire, C., del Mar Ferradás, M., Regueiro, B., Rodríguez, S., Valle, A., and Núñez, J. C. (2020). Coping strategies and self-efficacy in university students: a person-centered approach. *Front. Psychol.* 11, doi: 10.3389/fpsyg.2020.00841

García-Carmona, M., Marín, M. D., and Aguayo, R. (2019). Síndrome de agotamiento en profesores de secundaria: una revisión sistemática y metaanálisis. *Soc. Psychol. Educ.* 22, 189–208. doi: 10.1007/s11218-018-9471-9

Garrido, F. A. Z. (2020). Docência universitária durante a pandemia da COVID-19: um olhar do Chile. *Rev. Docência Ensino Superior* 10, 1–9. doi: 10.35699/2237-5864.2020.24777

Hobfoll, S. E. (1989). Conservation of resources: a new attempt at conceptualizing stress. *Am. Psychol.* 44 (3), 513–524. doi: 10.1037/0003-066X.44.3.513

Hu, L. T., and Bentler, P. M. (1999). Cutoff criteria for fit indexes in covariance structure analysis: conventional criteria versus new alternatives. *Struct. Equ. Model.* 6 (1), 1–55. doi: 10.1080/10705519909540118

Jin, B., and Kim, J. (2017). Grit, basic needs satisfaction, and subjective well-being. *J. Individ. Differ.* 38 (1), 29–35. doi: 10.1027/1614-0001/a000219

Keyes, C. L. M. (2002). The mental health continuum: from languishing to flourishing in life. *J. Health. Soc. Behav.* 43 (2), 207–222. doi: 10.2307/3090197

Kim, L. E., and Asbury, K. (2020). 'As if a rug had been pulled from under you': the impact of COVID-19 on teachers in England during the first six weeks of the UK lockdown. *Br. J. Educ. Psychol.* 90 (4), 1062–1083. doi: 10.1111/bjep.12381

Kock, N. (2015). Common method bias in PLS-SEM: A full collinearity assessment approach. *Int. J. e-Collab.* 11 (4), 1–10. doi: 10.4018/ijec.2015100101

König, J., Jäger-Biela, D. J., and Glutsch, N. (2020). Adapting to online teaching during COVID-19 school closure: teacher education and teacher competence effects among early career teachers in Germany. *Eur. J. Teach. Educ.* 43 (4), 608–622. doi: 10.1080/02619768.2020.1809650

Laca Arocena, F., Mejía Ceballos, J. C., Yáñez Velasco, C. L., and Mayoral Sánchez, E. G. (2011). Factores afectivos y de personalidad como antecedentes del bienestar laboral en profesores españoles y mexicanos. *Pensamiento Psicológico* 9 (16), 11–26.

Lücker, P., Kästner, A., Hannich, A., Schmeyers, L., Lücker, J., and Hoffmann, W. (2022). Stress, coping and considerations of leaving the profession. *Int. J. Environ. Res. Public. Health.* 19 (23), 16122. doi: 10.3390/ijerph192316122

Madigan, D. J., and Kim, L. E. (2021). Towards an understanding of teacher attrition: a meta-analysis of burnout, job satisfaction, and teachers' intentions to quit. *Teach. Teach. Educ.* 105, 103425. doi: 10.1016/j.tate.2021.103425

Marpa, E. P. (2021). Technology in the teaching of mathematics: an analysis of Teachers' attitudes during the COVID-19 pandemic. *Int. J. Stud. Educ.* 3 (2), 92–102. doi: 10.46328/ijonse.36

Maxwell, S. E., and Cole, D. A. (2007). Bias in cross-sectional analyses of longitudinal mediation. *Psychol. Methods.* 12 (1), 23–44. doi: 10.1037/1082-989X.12.1.23

McLay, K. F., Thomasse, L., and Reyes, V. C. (2023). Embracing discomfort in active learning and technology-rich higher education: insights through reflective inquiry. *Educ. Technol. Res. Dev.* 71, 1161–1177. doi: 10.1007/s11423-023-10192-6

Morales-Loor, K. P., Romero-Amores, N. V., Bayas-Jaramillo, C. M., and Vasco-Delgado, J. C. (2025). Integración de la tecnología en la formación docente. *MLAJ* 3 (1), 448–467. doi: 10.62131/MLAJ-V3-N1-022

Muñoz-Campos, E., Fernández-González, A., and Jacott, L. (2018). Bienestar subjetivo y satisfacción vital del profesorado. *REICE. Revista Iberoamericana sobre Calidad, Eficacia y Cambio en Educación*. 16 (1), 105–117. doi: 10.15366/reice2018.16.1.007

Navarro-Montaño, M. J., Piñero-Virué, R., Jiménez Navarro, P., and Mateo Navarro, V. (2022). Estrategias didácticas con TIC en la formación del profesorado de educación Primaria. *Ciencia Educ.* 6 (3), 21–33. doi: 10.22206/cyed.2022.v6i3

Nwoko, J. C., Emeto, T. I., Malau-Aduli, A. E. O., and Malau-Aduli, B. S. (2023). Una revision sistemática de los factores que influyen en el bienestar laboral de los docentes. *Rev. Int. Investig. Ambiental Salud Públ.* 20 (12), 6070. doi: 10.3390/ijerph20126070

Podsakoff, P. M., MacKenzie, S. B., Lee, J.-Y., and Podsakoff, N. P. (2003). Common method biases in behavioral research: a critical review of the literature and recommended remedies. *J. Appl. Psychol.* 88 (5), 879–903. doi: 10.1037/0021-9010.88.5.879

Rodríguez-Fuentes, B., Sepúlveda-Páez, G., and Valenzuela-Jofré, C. (2023). Experiencia de prácticas profesionales online de estudiantes de Psicología de la Universidad de Talca, Chile. *Rev. Sul-Am. Psicol.* 10 (2), 191–214. doi: 10.29344/2318650x.2.3232

Rögele, S., Rilling, B., Apfel, D., and Fuchs, J. (2022). Sustainable development competencies and student-centered teaching strategies. *IJSHE* 23 (6), doi: 10.1108/IJSHE-02-2021-0069

Ryff, C. D. (1989). Happiness is everything, or is it? Explorations on the meaning of psychological well-being. *J. Pers. Soc. Psychol.* 57 (6), 1069–1081. doi: 10.1037/0022-3514.57.6.1069

Schetsche, C., Gago-Galvagno, L., and Mustaca, A. E. (2023). Estrategias de regulación emocional. *Rev. CES Psicol.* 16, 1. doi: 10.21615/cesp.6281

Shvay, R., Petrovska, I., Kushka, B., Sydorovych, O., and Mys'kiv, I. (2025). The relationship between teachers' psychological well-being, coping strategies and intolerance of uncertainty: a comparative study in Poland and Ukraine. *Eur. J. Educ. Res.* 14 (3), 1031–1045. doi: 10.12973/eu-jer.14.3.1031

Silva-Domínguez, D., and Luque-Villanueva, Y. (2022). Miedo al COVID-19, ansiedad, depresión y satisfacción familiar durante la pandemia COVID-19 en adultos de Lima Metropolitana. *CASUS. Revista de Investigación y Casos en Salud1.* 6 (1).

Sokal, L., Trudel, L. E., and Babb, J. (2021). I've had it! factors associated with burnout and low organizational commitment in Canadian teachers during the second wave of the COVID-19 pandemic. *Int. J. Educ. Res. Open* 2, 100023. doi: 10.1016/j.ijedro.2020.100023

Sonnenberg, D., and Rutledge, P. (2024). Online education benefits instructors' emotional labor management. *Comput. Educ. Open* 7, 100225. doi: 10.1016/j.caeo.2024.100225

Soyer, G. F. (2019). Urie bronfenbrenner: the ecology of human development book review. *J. Cult. Values Educ.* 2 (2), 77–80. doi: 10.46303/jcve.02.02.6

Stockinger, K., Daumiller, M., and Dresel, M. (2025). Relations between university teachers' teaching- related coping strategies and well-being over time: a cross-lagged panel analysis. *Br. J. Educ. Psychol.* [Advance online publication]., doi: 10.1111/bjep.12777

Varela, J., Guzmán, P., Oriol, X., Romo, F., and Miranda, R. (2023). Bienestar, afectos y desgaste laboral de los profesores durante la pandemia en Chile. *Rev. Psicodidáctica* 28 (1), 10–18. doi: 10.1016/j.psicod.2022.07.002

Vargas Rubilar, N., and Oros, L. B. (2021). Stress and burnout in teachers during times of pandemic. *Front. Psychol.* 12, 756007. doi: 10.3389/fpsyg.2021.756007

Wang, H., Sun, W., Zhou, Y., Li, T., and Zhou, P. (2022). Teachers' assessment literacy improves teaching efficacy. *Front. Psychol.* 13, 1007830. doi: 10.3389/fpsyg.2022.1007830

Yurt, E. (2022). Collective teacher self-efficacy and burnout. *Int. J. Modern Educ. Stud.* 6 (1). doi: 10.51383/ijonmes.2022.168